



A new formulation for predicting the ultimate capacities of FRP-confined concrete using advanced machine learning framework: Developed structural reliability analysis

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ARTICLE INFO

Keywords:

Fibre reinforced polymer
Confined concrete
Ultimate capacities
Machine learning
Group Method Data Handling Neural Network
Reliability Analysis

ABSTRACT

Maintaining the safety levels of concrete structures is critical, where boosting the strength and deformability of such components utilizing Fibre Reinforced Polymer (FRP) composites is well adopted. However, the outside environment and applied loads can influence the FRP confined concrete. Therefore, accurate assessment of the ultimate capacities of FRP-confined concrete is highly essential. This paper proposes an advanced machine-learning framework using a novel approach called Group Method Data Handling Neural Network (GMDH-NN) for developing novel formulations that properly predict the ultimate capacities of FRP-confined concrete. The GMDH-NN ability in terms of tendency, agreement, and accuracy is compared to the existing empirical correlations and well-known machine-learning methodologies, notably Artificial Neural Network (ANN) and Response Surface Method (RSM). Furthermore, to address the failure modes of strength and ductility, the reliability analysis of FRP-confined concrete using combined Monte Carlo simulation (MCS) with the proposed GMDH-NN is examined for three types of composite sheets: Aramid, Carbon, and Glass. Results prove the GMDH-NN out-performance for modeling the ultimate strength ($R^2 = 0.871$) and strain ($R^2 = 0.79$) capacities for FRP-confined concrete. The structural reliability analysis indicated that the choice of FRP sheets had a substantial influence on the increase of the safety levels of confined concrete cores, as well as the requirement for early prediction of these parameters to avoid failures.

1. Introduction

Fiber-reinforced polymer (FRP)-confined columns (e.g., using Carbon, Glass or Aramid FRP) demonstrate high resistance and acceptable retrofitting conditions under axial stresses caused by the external environmental uncertainties such as wind or earthquake [1,2]. Under applied axial loads, the stress-strength and strain-ductility capacities of the columns warped by FRP sheets tend to increase [3–5]. These capacities are calculated using various parameters, which are subjected to uncertainties, including the mechanical properties (e.g., elastic modulus of FRP, actual hoop rupture strain of FRP, and

unconfined compressive strength of concrete), segment size (e.g., FRP sheet thickness, cross-sectional concrete cores), and the applied stress and strain loads [6–10]. Therefore, assessing the resistance of FRP-confined columns requires a trustworthy model that accounts for associated uncertainties. The design-oriented models are typically well-used to approximate the strength and strain capacities of FRP-confined segments [11–16]. Various studies have been conducted to evaluate the performances of FRP-confined concrete under improved mechanical characteristics in terms of strength and strain, precisely using closed form design-oriented models [17,18]. However, because of the complexity of the modelling process for these characteristics,

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<https://doi.org/10.1016/j.istruc.2024.107158>

Received 9 April 2023; Received in revised form 31 July 2024; Accepted 23 August 2024

Available online 10 September 2024

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developing an accurate design model for determining the ultimate capacities of FRP-confined columns is seen as a considerable challenge, particularly when conducting reliability analysis. This latter is typically used to analyze the resilience state of reinforced FRP columns based on two failure modes: strength and ductility, in which precise models are required for accurate failure probability calculation. Thus, offering accurate models for predicting the behavior of ultimate capacities of FRP-confined concrete plays an important role in improving the resilience of such structures.

Over the past decades, various empirical models were developed to approximate the ultimate capacities of FRP-confined concrete segments. These correlations were designed using various input variables derived from the physical characteristics and composite core sizes. Almost all these models are established using the lateral confining pressure, which is governed by the mechanical characteristics and FRP sizes i.e. $\frac{2E_f t_f}{D} \varepsilon_{h,rupt}$ (where, t_f = FRP thickness, E_f = FRP elastic modulus, D = diameter of concrete core and $\varepsilon_{h,rupt}$ = FRP hoop rupture strain of the wrap) [4, 19–21]. The ultimate capacities of FRP-confined concrete cores have been modeled using two primary factors based on the confinement stiffness ratio i.e. $\frac{2E_f t_f}{D(f_{co}/\varepsilon_{co})}$ and strain ratio i.e. $\frac{\varepsilon_{h,rupt}}{\varepsilon_{co}}$ (where, ε_{co} = unconfined concrete strain and f_{co} = unconfined concrete stress), which are extended in refs. [7,8,16,22]. Ozbakkaloglu and Lim [11] established empirical models based on the lateral confining pressure and lateral stiffness i.e. $K_f = 2E_f t_f / D$. According to their findings, multi-source uncertainties are recorded during the prediction process in the form of performance functions, which can be utilized to approximate the failure domain of FRP-warped concrete cores. Keshtegar et al. [17] applied the first order reliability method (FORM) based-conjugate sensitivity vector for the concrete cores' reliability analysis using the ultimate capacities based performance functions. The failure zones in their reliability study were determined using the empirical models established earlier by Ozbakkaloglu and Lim [11]. Their study revealed that the primary failure modes for deformed FRP concrete cores include strength and ductility, whereas during the reliability analysis two probabilistic ultimate stain and strength capacities are used to approximate the failure domain of the concert core. However, results suggested to improve the accuracy of the empirical models in order to produce reliable and trustworthy designs. Therefore, one of the primary difficulties in probabilistic analysis of FRP-confined concrete segments is the accuracy of anticipated models for estimating the failure domain. Consequently, there is a gap in the appropriate approximation of the failure probability for this dimensional reliability problem, which includes two failure modes and associated multi-source uncertainties in material properties and composite core sizes. As a result, the scientific community looked for more advanced approaches to overcome these shortcomings and produce more accurate models.

Recently, researchers adopted the use of machine learning (ML) based predictive models as powerful soft computing techniques to solve various types of complex problem in civil and structural engineering [23,24]. These techniques, also known as data-driven models, have recently been used to estimate the ultimate strength and strain capacities of FRP-reinforced concrete columns [25–35]. The first correlation based ML-predictive model to describe the ultimate capacities of concrete cores warped by FRP was generated utilizing the genetic programming (GP) [25]. The obtained findings utilizing the GP-based model demonstrated a clear outperformance in terms of prediction compared to the empirical relations. Since then, different ML techniques have been employed and compared to address the complex behavior of the ultimate capacities of FRP-confined concrete. Cevik et al. [36] proposed a combined GP and stepwise regression (SR) framework for CFRP-confined concrete cylinders. This latter were evaluated and compared to two machine learning approaches: neuro-fuzzy (NF) and artificial neural networks (ANN) in order to reinforce the prediction accuracy [34]. The study findings confirmed the superiority and accuracy of ML-based methodologies compared to the employed empirical

correlation, whereas the acquired results suggested better accuracy in estimating the ultimate strength capacity rather than the ultimate strain capacity. In another study, using a database composed of 95 experimental tests, the performance of ANN, Group method of data handling (GMDH) and GP were compared in order to predict the ultimate strength of reinforced concrete columns warped with FRP [26]. For this particular case, ANN predictions outperformed the other models in terms of accuracy. Furthermore, the compressive strength of FRP-confined concrete was estimated using several ANN models, and the supplied prediction results revealed a better approximation to solve the problem when compared to existing empirical correlations [28,33,35,37,38]. Chen et al. [39] compared three ML-models, including ANN with back-propagation training approach, GP and support vector machine (SVM) for predicting the ultimate axial strain of FRP-confined concrete cylinders. SVM predictions were shown to be the most accurate when compared to other ML models and empirical correlations. Similarly, Mansouri et al. [32] used the radial basis neural network (RBNN), adaptive neuro fuzzy inference system (ANFIS) and M5 model tree (M5Tree) to predict the ultimate strength and strain capacities of concrete cylinders warped with FRP sheets. The results showed that the RBNN and ANFIS models performed similarly, whereas the M5Tree model fared poorly. Ben Seghier et al. [40] took a new approach to the problem, connecting the ML model with meta-heuristic optimization methods to increase its performance. The hybrid machine learning models used support vector regression (SVR) as the basic model with genetic, particle swarm, and whale optimization methods as meta-heuristic optimization algorithms. The results revealed that hybrid SVR models outperform SVR-standalone with high accuracy and low uncertainty. Recently, Keshtegar et al. [10] suggested a novel hybrid SVR paired with the response surface method (RSM) for solving this problem, and the results show that the proposed model outperforms SVR, RSM standalone, and empirical correlations.

To summarize, three major challenges surround the use of ML to tackle the current problem of accurately estimating the ultimate capacities of FRP-confined concretes: i) Developing accurate prediction models with low uncertainties and a closed form empirical formulation for modeling strain and strength behaviors; ii) These innovative models should be built on highly flexible nonlinear functions obtained from a variety of ML methodologies; and iii) A lack of contribution to the structural reliability analysis of composite concrete columns with multi-failure modes and varied uncertainties arising from the interaction of FRP and concrete composites. As a result, more accurate ML models should be proposed to estimate the system failure probability utilizing the ultimate condition of concrete warped by FRP.

Based on the foregoing justifications and restrictions, the goal of this study is to advance the existing state of the art by proposing novel formulations for the ultimate capacities (strain and strength) of FRP-confined concrete using a novel ML-based framework at stage one. The proposed ML framework introduces the GMDH-based NN (GMDH-NN) technique as a novel approach to accurately anticipate the ultimate capacities for composite concrete cores. In addition, two other ML models, ANN and RSM, as well as the existing empirical correlation, are proposed for comparison purposes. All ML-predictive models are trained and tested on a broad and diverse experimental database containing over 750 segments. Following that, the models' findings are submitted to various performance evaluations using several metrics and graded based on accuracy and efficiency. The second stage includes a rigorous reliability analysis to assess the safety levels of FRP-confined concrete with different warped sheets. The analysis focuses on two types of failure modes: strength and ductility, as well as three types of sheets: CFRP, GFRP, and AFRP. The MCS is integrated with the GMDH-NN to produce a unique surrogate model for approximating failure domain-based probabilistic performance functions. Finally, the reliability analysis results are discussed and contrasted to the failure probability findings.

2. Brief overview on the previously established correlations

The lateral interactions of confining pressure between FRP sheets and concrete core have a substantial impact on the performance of composite segments, represented by the ultimate capacities in terms strength (f'_{cc}) and strain (ϵ_{cu}) [41]. Therefore, the real ultimate confining pressure at FRP rupture ($f_{lu,a}$) is used as the primary parameter to determine the ultimate capacities of FRP-confined concrete [42]. This parameter is derived by assuming evenly distributed stress around the circumference as expressed in the following [43]:

$$f_{lu,a} = \frac{2E_f t_f}{D} \epsilon_{h,rupt} \quad (1)$$

where, $\epsilon_{h,rupt}$ is the hoop rupture strain of the FRP sheet, t_f is the FRP thickness of FRP, E_f is the FRP elastic modulus and D is the concrete core diameter. According to the plane stress principal of a concrete section subjected to a lateral confining pressure under axial core compression, the tension stress exists in the FRP sheet along the hoop direction, and therefore the FRP sheets are ruptured at $\epsilon_{h,rupt}$. In addition, Eq. (1) for the ultimate lateral pressure is obtained using the failure model-based FRP deformation theory. Thus, the existing empirical correlations derived from the ultimate lateral pressure and utilized to predict f'_{cc} and ϵ_{cu} of FRP-confined concrete can be ben reported as:

i) Toutanji (1999) model [9].

$$\frac{f'_{cc}}{f'_{co}} = 1 + 3.5 \left(\frac{f_{lu,a}}{f'_{co}} \right)^{0.85} \quad (2)$$

$$\frac{\epsilon_{cu}}{\epsilon_{co}} = 1 + \left(310.57 \epsilon_{h,rupt} + 1.9 \right) \left(\frac{f'_{cc}}{f'_{co}} - 1 \right) \quad (3)$$

where, ϵ_{co} is the corresponding strain to unconfined concrete strength (f'_{co}). The confinement stiffness ratio (ρ_K) and strain ratio (ρ_ϵ) are two factors that used to calculate $f_{lu,a}$ i.e. $f_{lu,a} = \rho_K \rho_\epsilon f'_{co}$ [8]. As a result, we obtain:

$$\rho_K = \frac{2E_f t_f}{D(f'_{co} / \epsilon_{co})} \quad (4)$$

$$\rho_\epsilon = \frac{\epsilon_{h,rupt}}{\epsilon_{co}} \quad (5)$$

Several empirical correlations are produced by using the aforementioned two parameters (Eqs. (4) and (5)), where two model categories are presented and employed for the prediction as follows:

ii) Teng et al. (2009) model [8].

$$\frac{f'_{cc}}{f'_{co}} = 1 + 3.5 \left(\rho_K - 0.01 \right) \rho_\epsilon \quad (6)$$

$$\frac{\epsilon_{cu}}{\epsilon_{co}} = 1.75 + 6.5 \rho_K^{0.8} \rho_\epsilon^{1.45} \quad (7)$$

iii) Sadeghian and Fam (2015) model [7].

$$\frac{f'_{cc}}{f'_{co}} = 1 + \left(2.77 \rho_K^{0.77} - 0.07 \right) \rho_\epsilon^{0.91} \quad (8)$$

$$\frac{\epsilon_{cu}}{\epsilon_{co}} = 1.5 + 6.78 \rho_K^{0.63} \rho_\epsilon^{1.08} \quad (9)$$

To estimate the ultimate strength and strain of FRP-confined concrete under axial load, a hybrid format of these models is extended by utilizing both $f_{lu,a}$ and factors ρ_K , ρ_ϵ . Using the three components, two models are established previously as:

iv) Ozbakkaloglu and Lim (2013) model [11].

$$\frac{f'_{cc}}{f'_{co}} = 1 + 0.0058 \frac{K_l}{f'_{co}} + 3.22 \left(\frac{f_{lu,a}}{f'_{co}} - \frac{f_{lo}}{f'_{co}} \right) \quad (10)$$

$$\frac{\epsilon_{cu}}{\epsilon_{co}} = 2 - \frac{f'_{co} - 20}{100} + 0.271 \left(\frac{K_l}{f'_{co}} \right)^{0.9} \frac{\epsilon_{h,rupt}^{1.35}}{\epsilon_{co}} \quad (11)$$

with,

$$f_{lo} = K_l \left(0.43 + 0.009 \frac{K_l}{f'_{co}} \right) \epsilon_{co} \quad (12)$$

where $K_l = 2E_f t_f / D$ for $K_l > f_{co}^{1.65}$ is a factor defined in the above formula for computing the ultimate capacities.

v) Keshtegar et al. (2017) model [16].

$$\frac{f'_{cc}}{f'_{co}} = 1 + (3.23 - 4.8 \rho_K^{2.5}) \left(\frac{f_{lu,a}}{f'_{co}} \right)^{0.95} \quad (13)$$

$$\frac{\epsilon_{cu}}{\epsilon_{co}} = 1 + (7.31 + 2.06 \rho_\epsilon^{0.8}) \left(\frac{f_{lu,a}}{f'_{co}} \right)^{0.6} \quad (14)$$

As it can be observed, the nonlinearity degree corresponding to the $f_{lu,a}/f'_{co}$ factor supplied by the empirical models in Eqs. (13) and (14) for predicting $\epsilon_{cu}/\epsilon_{co}$ is higher than the nonlinearity degree related to f'_{cc}/f'_{co} . As a result, it is clear that advanced fixable nonlinear models should be used for reliable prediction of $\epsilon_{cu}/\epsilon_{co}$.

3. Proposed machine learning framework

The proposed GMDH-NN approach is developed in the current study to establish an accurate formulation for the prediction of the ultimate capacities of FRP-confined concrete. The model structure is established based on the polynomial function as well as the response surface method, while a network of nodes predicted using RSM with two inputs are employed to build the final output model. The predicted data sets applied in the hidden nodes are computed using a second-order polynomial function with a cross term, where connection between the input layer and output layer is similar to the neural network formwork. In the following subsection, the three machine learning given from ANN, RSM and GMDH-NN are briefly introduced for modelling the ultimate stain and strength capacities of FRP-confined concrete cores.

3.1. Artificial neural network (ANN)

The key effort in applying machine learning-based ANN for the prediction of ultimate axial strength and strain is to connect the input variables $X \in (D, t, f'_{co}, \epsilon_{h,rupt}, E_f)$ into the output $Y \in (f'_{cc}/f'_{co}, \epsilon_{cc}/\epsilon_{co})$. As shown in Fig. 1, the ANN base multi-layer network provides a nonlinear predictive model with three fundamental layers as input layer, hidden layer, and output layer [44]. The mathematical formula for estimating a

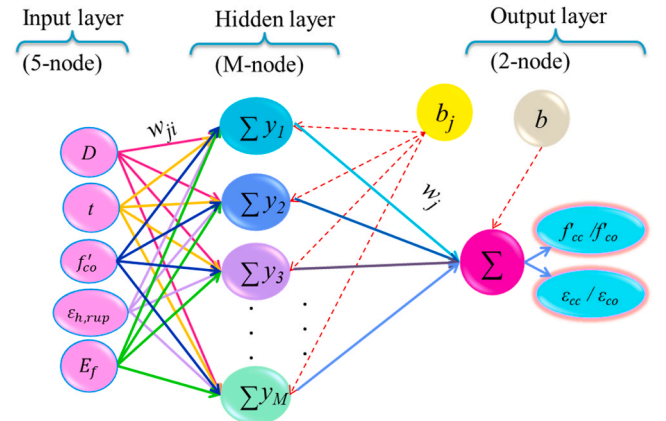


Fig. 1. The schematic structure of ANN with M-hidden node.

response using the network shown in Fig. 1 is represented in Eq. (15):

$$y = b + \sum_{j=1}^M \frac{w_j}{1 + \exp \left[\sum_{i=1}^n -b_j - w_{ij}x_i \right]} \quad (15)$$

where, b and, b_j are respectively the biases for the output and the i -th hidden nodes. w_j and w_{ji} are respectively the weights for the j -th hidden nodes. w_j is used to joint the output and the hidden nodes, and the i -th input node, while w_{ji} is applied to join the input nodes to the hidden nodes. M and n are respectively related to the number of the hidden nodes and the number of input variables.

In ANN model, the training technique for establishing biases and weights is a critical challenge. A suitable model with a robust and high-tendency relationship may be presented for accurate estimation of the unknown coefficients, including biases and weights. The back-propagation (B.P) method is a typical ANN training methodology used to calibrate the nonlinear relationship seen in Eq. (15)[45]. In this work, the B.P employing the Levenberg-Marquardt method is used to train the ANN model due to its efficiency and global convergence [46,47].

3.2. Response surface methods (RSM)

The RSM is a simple and effective surrogate model trained using a regression technique such as polynomial basis functions. The regression function of the applied RSM model is based on the second-order polynomials with cross-terms defined by the following formula [48]:

$$y = w_0 + \sum_{i=1}^n w_i x_i + \sum_{i=1}^n \sum_{j=1}^n w_{ij} x_i x_j \quad (16)$$

where the weight variables w_i and w_{ij} are utilized as connection elements between the input and output variables. To calibrate the model in order to estimate the unknown coefficients using the input and output variables, the least square estimator is used. As a result, when compared to soft computing models such as the ANN model, the RSM may produce inaccurate predictions [49,50]. Consequently, to increase the flexibility for training highly nonlinear problems, the structure or regression function of RSM should be improved..

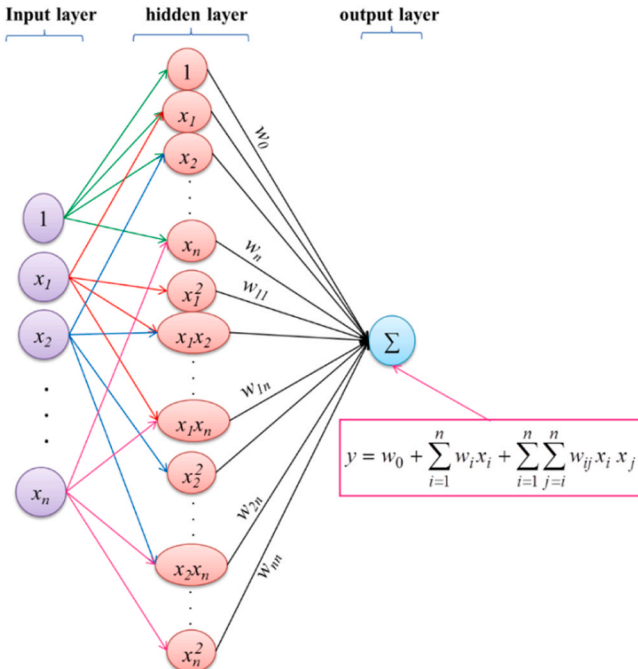


Fig. 2. The schematic structure of RSM.

3.3. Group Method of Data Handling-based neural network (GMDH-NN)

The GMDH-NN model is developed as a novel formulation for predicting the ultimate capacities of FRP-confined concrete cores. This model is provided by using RSM for selecting NN nodes in the modeling phase, while a network framework is used to anticipate the final strength and strain capacities. Five inputs variables including concrete core diameter (D), thickness of FRP (t) unconfined concrete strength (f'_{co}), actual hoop rupture strain of FRP ($\epsilon_{h,rupt}$) and FRP elastic modulus (E_f) should be linked to nonlinear responses for modeling the ultimate capacities as the ultimate strength ratio (f'_{cc}/f'_{co}) and ultimate strain ratio ($\epsilon_{cc}/\epsilon_{co}$). The GMDH-NN model is trained using a polynomial basis function with 2-order and one cross term [51]. In this method, the polynomial function is used as a regression function for each node, followed by two inputs as expressed in [52]:

$$y = a_0 + a_1 x_i + a_2 x_j + a_3 x_i^2 + a_4 x_j^2 + a_5 x_i x_j \quad (17)$$

where, a_0 to a_5 are unknown coefficients representing weight connections between the two inputs x_i, x_j and the predicted data, $\mathbf{O} \in (f'_{cc}/f'_{co}, \epsilon_{cc}/\epsilon_{co})$. Thus, for a giving N train data points with multi-dimensional input ($x_1, x_2, \dots, x_n$), output data of $\mathbf{O}$ and pairs (x_i, x_j), The coefficients, $a_0 - a_5$ are determined by least square estimator, given as follows [53]:

$$\mathbf{a} = [\mathbf{A}^T \mathbf{A}]^{-1} \mathbf{A}^T \mathbf{O} \quad (18)$$

where, $\mathbf{a} = \begin{Bmatrix} a_0 \\ a_1 \\ a_2 \\ a_3 \\ a_4 \\ a_5 \end{Bmatrix}$, $\mathbf{A} = \begin{bmatrix} 1 & x_{i1} & x_{j1} & x_{i1}^2 & x_{j1}^2 & x_{i1}x_{j1} \\ 1 & x_{i2} & x_{j2} & x_{i2}^2 & x_{j2}^2 & x_{i2}x_{j2} \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ 1 & x_{iN} & x_{jN} & x_{iN}^2 & x_{jN}^2 & x_{iN}x_{jN} \end{bmatrix}$

and $\mathbf{O} = \begin{Bmatrix} O_1 \\ O_2 \\ \vdots \\ O_N \end{Bmatrix}$.

The predicted data for each node is computed using this simple and straightforward formulation, which is given in Eq. (18), for each prior set of inputs. It is more vital to establish a link between the input variables and the ultimate capacities responses. The proposed GMDH-NN framework is shown in Fig. 3, in which nodes in the hidden layers are chosen based on the expected relationship in Eq. (18) and prior inputs from the previous layer. As shown in Fig. 3, the GMDH includes numerous hidden layers with y_{pM} elements, $p = 1, 2, \dots, p$ and $M = 1, 2, \dots, M$, where, p and M refer to the number of hidden layers and nodes in layers, respectively. As a result, in order to determine the nodes in the next layer, the predicted data for each node is generated using the two nodes from the previous levels. This model offers a straightforward training technique, a quick calibration process, and a high degree of nonlinearity.

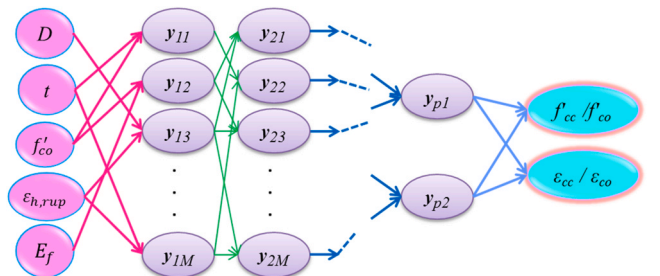


Fig. 3. The schematic structure of proposed GMDH-NN.

4. Data and statistical indicators

4.1. Experimental database

Establishing a novel formulation for predicting the ultimate capacities of FRP-confined concrete based on the proposed GMDH-NN approach and/or other machine learning techniques (i.e. ANN and RSM) necessitates a comprehensive descriptive database of real experimental testing. Over 738 experimental test samples were acquired from open research studies in the literatures to attain this purpose [7,16,22]. The utilized database in this study is not only for the purpose of developing novel formulations, but also for the performance evaluation of the existing well-known correlations that characterize the strain and strength capacities for FRP-confined concretes. The descriptive statistical details of the employed database are reported in Table 1. According to this table, the database is randomly divided into two parts: training with 590 sample tests (80%) and testing with 148 sample tests (20%). The first portion of the database is used to train – construct – the machine learning based models, while the second portion is utilized to test the performance of the developed models. Furthermore, the database contains seven variables, five of which are used as input variables, including concrete diameter, D (mm); thickness of FRP jacket, t (mm);

unconfined concrete strength, f'_{co} (MPa); actual hoop rupture strain of FRP, $\varepsilon_{h,rupt}$ (%) and the elastic modulus, E_f (GPa). The desired outputs, on the other hand, are the ultimate strain ratio, $\varepsilon_{cc} / \varepsilon_{co}$, and the ultimate strength ratio, f'_{cc} / f'_{co} .

Fig. 4 shows dataset distribution using histograms in the training and testing phases for f'_{cc}/f'_{co} and $\varepsilon_{cu}/\varepsilon_{co}$. It can be seen that the distribution of f'_{cc}/f'_{co} has a high-frequency in the range of 1.4–2.4, while it increases for $\varepsilon_{cu}/\varepsilon_{co}$ in the range of 2.5–11. This indicates that the ultimate strain ratio $\varepsilon_{cu}/\varepsilon_{co}$ shows a larger variety and disperse datasets leading to higher uncertainty compared to the ultimate strength capacity ratio, f'_{cc}/f'_{co} . Therefore, the extended model for the ultimate strain ratio should be provided with a high flexibility for nonlinearly degree in order to provide an appropriate prediction result. However, the ultimate strength ratio is trained based on a moderate nonlinear function derived from the investigated machine learning methodologies. On the other hand, the data distributions in both phases report large varieties with different distributions that the modes for training data in f'_{cc}/f'_{co} and $\varepsilon_{cu}/\varepsilon_{co}$ are respectively obtained about 1.9 (rounded to 2) and 6.8 (rounded to 7), which is differed to the testing datasets e.g. 1.7 for f'_{cc}/f'_{co} for and 7.5 for $\varepsilon_{cu}/\varepsilon_{co}$. This indicates the difference in the data used for training and testing, ensuring an accurate evaluation of the model's performance.

Table 1

Descriptive statistics of the experimental database used in the modeling process during the training and testing phases.

variables	Training datasets						Testing datasets					
	X_{min}	X_{max}	Mean	STD	COV	Skewness	X_{min}	X_{max}	Mean	STD	COV	Skewness
D (mm)	51	406	152.966	46.330	0.303	2.864	51	406	158.378	53.428	0.337	2.952
t (mm)	0.090	7.260	0.826	0.904	1.095	2.567	0.110	5.840	0.784	0.966	1.232	2.588
f'_{co} (MPa)	19.700	188.2	52.226	29.287	0.561	1.710	16.6	188.2	48.405	28.348	0.586	1.943
$\varepsilon_{h,rupt}$ (%)	0.110	4.980	1.232	0.565	0.459	1.080	0.120	3.090	1.256	0.584	0.465	0.505
E_f (GPa)	10.5	662.5	180.544	115.740	0.641	0.817	18.48	640	192.035	122.206	0.636	0.931
f'_{cc} / f'_{co}	1.100	6.830	2.003	0.790	0.394	2.316	1.100	5.360	2.079	0.733	0.353	1.673
$\varepsilon_{cc} / \varepsilon_{co}$	1.060	27.997	7.981	4.592	0.575	1.244	1.217	25.267	8.301	4.780	0.576	1.096

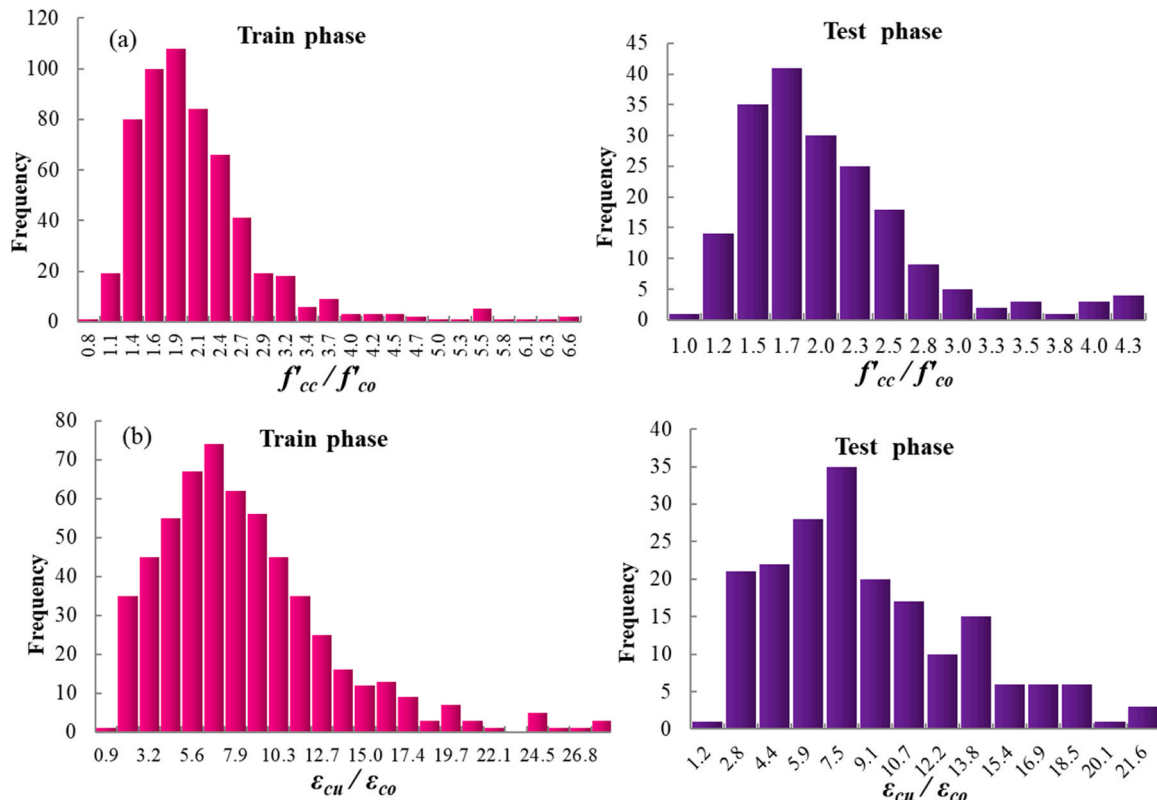


Fig. 4. Histograms of the output data for the train and test phases (a) f'_{cc}/f'_{co} and (b) $\varepsilon_{cu}/\varepsilon_{co}$.

4.2. Comparative statistical indicators

Several statistical indicators are used to assess the outcomes of the empirical correlation and the proposed machine learning methodologies, including the RSM, ANN, and the novel proposed approach, GMDH-NN. The 10 used indicators are expressed using Eqs.(19) to (28) as follows [54,55]:

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (E_i - P_i)^2} \quad (19)$$

$$MBE = \frac{1}{n} \sum_{i=1}^n E_i - P_i \quad (20)$$

$$MAE = \frac{1}{n} \sum_{i=1}^n |E_i - P_i| \quad (21)$$

$$MSE = \frac{1}{n} \sum_{i=1}^n (E_i - P_i)^2 \quad (22)$$

$$NSE = 1 - \frac{\sum_{i=1}^n (E_i - P_i)^2}{\sum_{i=1}^n (E_i - \bar{E})^2} \quad -\infty \leq NSE \leq 1 \quad (23)$$

$$d = 1 - \frac{\sum_{i=1}^n (E_i - P_i)^2}{\sum_{i=1}^n (|P_i - \bar{E}| + |E_i - \bar{E}|)^2} \quad 0 \leq d \leq 1 \quad (24)$$

$$CI = d \times NSE \quad (25)$$

$$R^2 = 1 - \frac{\sum_{i=1}^n (E_i - P_i)^2}{\sum_{i=1}^n (\bar{E} - P_i)^2} \quad (26)$$

$$Max(RE) = \max \left(\frac{E_i - P_i}{E_i} \times 100 \quad i = 1, 2, \dots, N \right) \quad (27)$$

$$Min(RE) = \min \left(\frac{E_i - P_i}{E_i} \times 100 \quad i = 1, 2, \dots, N \right) \quad (28)$$

where *RMSE* is the root mean square error; *MBE* is the mean bias error;

MAE is the mean absolute error; *MSE* is the mean square error, the best model among the others is the one with the lowest values of the preceding indexes. In the other hand, *d* is the agreement index; *NSE* is the Nash and Sutcliffe efficiency, *CI* is the confidence index and *R²* is the coefficient of determination. The top performing models, depending on these metrics, are those having the highest values among the rest. *Max (RE)* and *Min (RE)* are the maximum and the minimum relative error of the prediction given by model, respectively. *E_i* and *P_i* are the *i*-th experimental and predicted values, respectively, while $\bar{E}$ is the average value. *N* is the sample size.

5. Implementation, results and discussion

5.1. General framework

5.1.1. Implementation

The primary goal of this research is to establish new formulations based on sophisticated machine learning framework, which will be used to analyze the structural reliability of FRP-confined concrete under strength and ductile failure modes. In the first phase, the proposed framework incorporates the novel machine learning technique, GMDH-NN to build new formulations for predicting the ultimate capacities of FRP-confined concrete. The reliability study of the FRP-confined concrete with different types of sheets is then examined in the second phase using the Monte Carlo Simulation and the newly developed GMDH-NN formulations. Fig. 5 depicts the general structure of the suggested framework in this study.

5.1.2. GMDH-NN based new correlations

The overall modeling phase findings for the ultimate strength ratio (f'_{cc}/f'_{co}) prediction using the proposed GMDH-NN present the below close form polynomial functions:

i) Models in the first-hidden layer

$$y_{11} = 1.1101 + 0.00551x_5 + 0.1747x_2 - 1.0433 \\ \times 10^{-5}x_5^2 - 0.016507x_2^2 + 0.002934x_5x_2$$

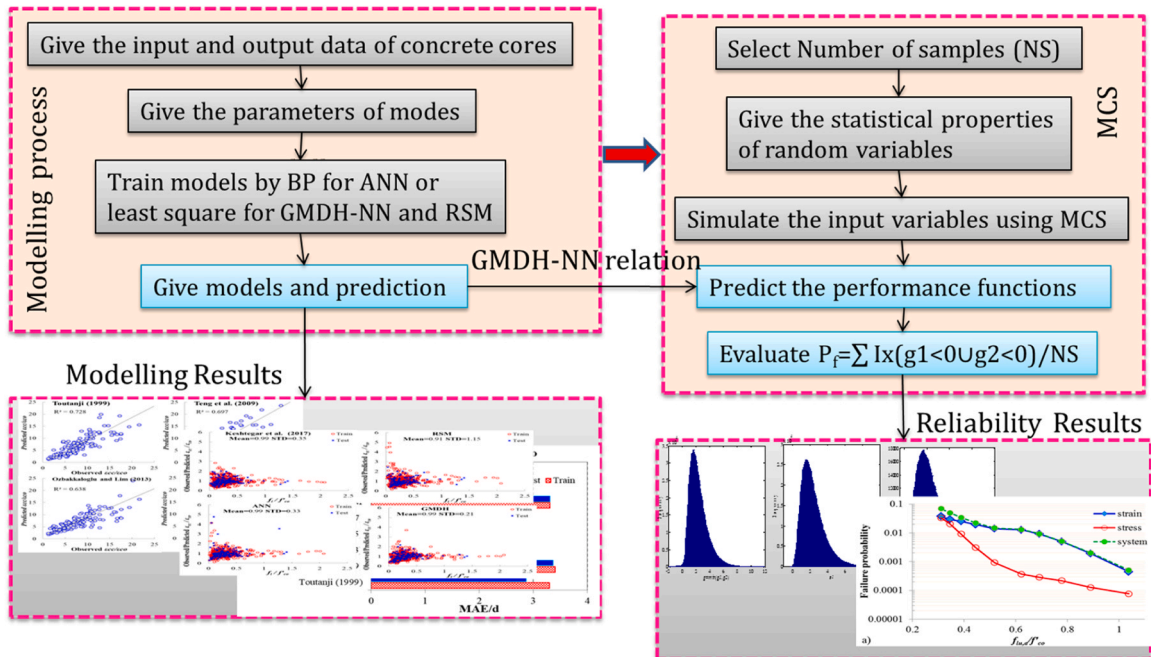


Fig. 5. Proposed machine learning and structural reliability analysis frameworks for simulating and estimating the failure domain of FRP-confined concrete.

$$y_{12} = 1.43774 + 0.55495x_4 + 0.24856x_2 - 0.088047x_4^2 \\ - 0.011837x_2^2 - 0.1231105x_4x_2$$

$$y_{13} = 2.93904 + 0.68257x_2 - 0.03534x_3 - 0.09569x_2^2 + 1.7253 \\ \times 10^{-4}x_3^2 - 0.002358x_3x_2$$

$$y_{14} = 3.99061 - 0.0079617x_1 - 0.03441x_3 + 1.0848 \\ \times 10^{-5}x_1^2 + 1.2733 \times 10^{-4}x_3^2 + 3.9064 \times 10^{-5}x_3x_1$$

ii) Models in the second-hidden layer

$$y_{21} = 9.76252 - 1.51809y_{11} - 8.73205y_{12} + 0.03419y_{11}^2 \\ + 1.90277y_{12}^2 + 1.23038y_{11}y_{12}$$

$$y_{22} = 2.08821 - 0.94907y_{11} - 1.37752x_4 + 0.302305y_{11}^2 \\ + 0.023077x_4^2 + 0.916115y_{11}x_4$$

$$y_{23} = 1.4717 - 0.5057y_{13} + 2.3402 \times 10^{-3}x_1 + 0.5843y_{13}^2 + 1.5668 \\ \times 10^{-5}x_1^2 - 5.2484 \times 10^{-3}y_{13}x_1$$

$$y_{24} = 7.1182 - 6.4861y_{14} - 0.05757x_4 + 1.81076y_{14}^2 \\ - 0.11121x_4^2 + 0.28686y_{14}x_4$$

iii) Models in the third-hidden layer

$$y_{31} = -0.2735 + 0.85841y_{21} + 0.44483y_{22} - 0.36586y_{21}^2 \\ + 0.0709y_{22}^2 + 0.21369y_{21}y_{22}$$

$$y_{32} = 6.19598 - 0.72675y_{23} - 4.560658y_{24} - 1.33055y_{23}^2 \\ - 0.739674y_{24}^2 + 3.6523y_{23}y_{24}$$

The final f_{cc}/f_{co} correlation based on GMDH-NN derived from the aforementioned predicted models is presented in Fig. 6 and determined as below:

$$f_{cc}/f_{co} = -0.84928 + 0.195076y_{31} + 0.53214y_{32} \\ - 0.12766y_{31}^2 - 0.200498y_{32}^2 + 0.6909y_{31}y_{32}$$

where,

$$x_1 = D(mm), \quad x_2 = t(mm), \quad x_3 = f_c(MPa), \quad x_4 = \varepsilon_{h, rup}, \quad x_5 = E(GPa).$$

Similarly to the ultimate strength ratio, the results of the modeling phase for the ultimate strain ratio ($\varepsilon_{cc} / \varepsilon_{co}$) prediction using GMDH-NN are presented in the below close form polynomial functions:

i) Models in the first-hidden layer

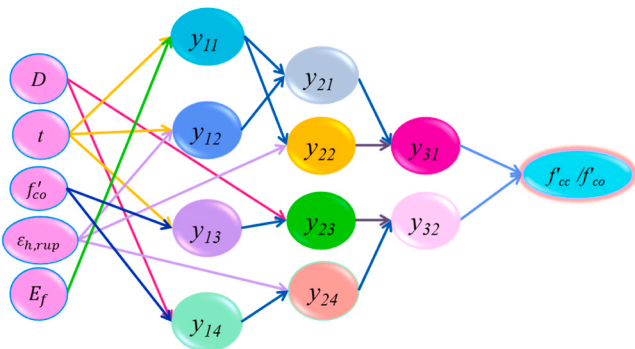


Fig. 6. GMDH-NN final model, including nodes, hidden layers and connections for f_{cc}/f_{co} prediction.

$$y_{11} = 11.3779 - 0.1547612x_3 + 0.022796x_1 + 5.68404 \\ \times 10^{-4}x_3^2 - 7.5268 \times 10^{-5}x_1^2 + 1.43361 \times 10^{-4}x_3x_1$$

$$y_{12} = 5.37215 - 0.10139x_3 + 6.88081x_4 + 4.64937 \\ \times 10^{-4}x_3^2 - 0.789512x_4^2 - 0.012421x_4x_3$$

$$y_{13} = 6.3218 + 0.01725x_5 + 0.02515x_2 - 5.14275 \\ \times 10^{-5}x_5^2 - 0.09761x_2^2 + 0.01198x_5x_2$$

$$y_{14} = 11.02198 + 0.019387x_5 - 0.09356x_3 - 4.4913 \\ \times 10^{-5}x_5^2 + 3.8196 \times 10^{-4}x_3^2 - 1.07296 \times 10^{-4}x_3x_5$$

$$y_{15} = 0.790284 + 1.623997x_2 + 6.88905x_4 - 0.187611x_2^2 \\ - 0.765478x_4^2 - 0.856104x_2x_4$$

$$y_{16} = 11.37794 - 0.154761x_3 + 0.022796x_1 + 5.684 \times 10^{-4}x_3^2 \\ - 7.5268 \times 10^{-5}x_1^2 + 1.4336 \times 10^{-4}x_3x_1$$

ii) Models in the second-hidden layer

$$y_{21} = 16.18435 + 1.2864x_2 - 4.049614y_{11} - 0.48072x_2^2 \\ + 0.34093y_{11}^2 + 0.190556y_{11}x_2$$

$$y_{22} = 9.941199 - 2.40801y_{12} - 1.21461y_{13} + 0.070833y_{12}^2 \\ + 0.051265y_{13}^2 + 0.28802y_{12}y_{13}$$

$$y_{23} = 3.216493 - 0.684922y_{14} - 0.158999y_{15} + 0.088671y_{14}^2 \\ + 0.0442382y_{15}^2 + 0.036695y_{14}y_{15}$$

$$y_{24} = 17.05658 + 1.58258 \times 10^{-3}x_5 - 3.63933y_{16} - 3.5712 \\ \times 10^{-5}x_5^2 + 0.30078y_{16}^2 + 9.29942 \times 10^{-4}y_{16}x_5$$

iii) Models in the third-hidden layer

$$y_{31} = 3.61255 - 0.51157y_{21} + 0.25131y_{22} + 0.047198y_{21}^2 + 0.02578y_{22}^2 \\ + 0.01929y_{21}y_{22}$$

$$y_{32} = 4.103541 + 0.046615y_{23} - 0.301073y_{24} - 2.31814 \\ \times 10^{-4}y_{23}^2 - 0.013118y_{24}^2 + 0.10047y_{23}y_{24}$$

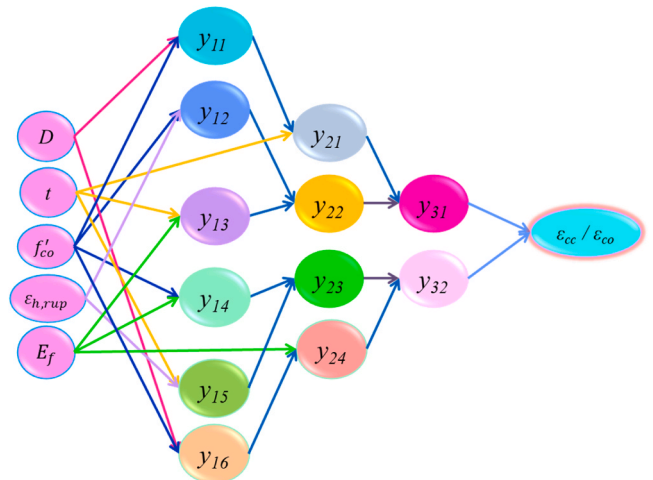


Fig. 7. GMDH-NN final model, including nodes, hidden layers and connections for $\varepsilon_{cc} / \varepsilon_{co}$ prediction.

The final ($\varepsilon_{cc} / \varepsilon_{co}$) correlation based on GMDH-NN derived from the aforementioned predicted models is presented in Fig. 7 and determined as below:

$$\varepsilon_{cc} / \varepsilon_{co} = -2.25225 + 0.8882y_{31} + 0.86747y_{32} - 0.149257y_{31}^2 - 0.32868y_{32}^2 + 0.42932y_{31}y_{32}$$

where,

$$x_1 = D(mm), \quad x_2 = t(mm), \quad x_3 = f'_c(MPa), \quad x_4 = \varepsilon_{h,rup}, \quad x_5 = E(GPa).$$

5.2. Framework evaluation and validation

5.2.1. Comparative performance evaluations

The statistical indicators in Section 4.2 are employed to evaluate and examine the performance of the five commonly used empirical correlations and the three machine learning algorithms, including RSM, ANN, and the proposed GMDH-NN methodology, in predicting the ultimate capacities of FRP-confined concrete. To begin, Table 2 and Table 3 summarize these performance evaluations. Based on the findings obtained from the tables, the performance of the GMDH-NN model is more accurate than the others by 4 out of 6 statistical metrics for the ultimate strength ratio (f'_{cc}/f'_{co}) and 5 out of 6 statistical metrics for the ultimate

Table 2

Comparative performance evaluation results, case 1: ultimate strength ratio (f'_{cc}/f'_{co}).

models		Toutanji (1999)	Teng et al. (2009)	Ozbakkaloglu and Lim (2013)	Sadeghian and Fam (2015)	Keshtegar et al. (2017)	RSM	ANN	GMDH-NN
Training phase	RMSE	0.4214	0.4094	0.3426	0.3359	0.3147	0.4638	0.3477	0.3044
	MSE	0.1776	0.1676	0.1174	0.1128	0.0991	0.2151	0.1209	0.0927
	MBE	0.1315	-0.0635	-0.0089	0.0194	0.0207	0.0347	0.0260	0.0266
	CI	0.6644	0.6833	0.7669	0.7792	0.8033	0.5797	0.7599	0.8151
	Max (RE)	1.1886	1.5388	2.0937	1.4465	1.4833	2.4947	1.6883	1.5108
	min (RE)	-3.4363	-4.0812	-0.9681	-2.4187	-1.0171	-1.5858	-0.8494	-0.7439
Testing phase	RMSE	0.3402	0.3761	0.3450	0.3078	0.3071	0.4399	0.3290	0.2647
	MSE	0.1157	0.1415	0.1190	0.0948	0.0943	0.1935	0.1082	0.0701
	MBE	0.0896	-0.1006	-0.0426	-0.0194	-0.0146	0.0229	-0.0152	0.0075
	CI	0.7404	0.6871	0.7295	0.7840	0.7838	0.5680	0.7513	0.8387
	Max (RE)	0.8970	1.1785	1.2782	1.0577	1.0588	1.5553	1.3714	0.8790
	min (RE)	-1.0179	-0.9549	-0.5646	-0.6685	-0.6761	-1.1889	-0.7540	-0.6183
Over-all	RMSE	0.4051	0.4028	0.3431	0.3303	0.3132	0.4590	0.3440	0.2965
	MSE	0.1652	0.1624	0.1177	0.1092	0.0981	0.2108	0.1184	0.0882
	MBE	0.1231	-0.0709	-0.0157	0.0117	0.0136	0.0323	0.0178	0.0227
	CI	0.6796	0.6841	0.7594	0.7802	0.7994	0.5773	0.7582	0.8198
	Max (RE)	1.1303	1.4667	1.9306	1.3688	1.3984	2.3068	1.6249	1.3845
	min (RE)	-2.9526	-3.4559	-0.8874	-2.0687	-0.9489	-1.5064	-0.8303	-0.7188

*The best outcomes are indicated in bold numbers.

Table 3

Comparative performance evaluation results, case 2: ultimate strain ratio ($\varepsilon_{cc} / \varepsilon_{co}$).

models		Toutanji (1999)	Teng et al. (2009)	Ozbakkaloglu and Lim (2013)	Sadeghian and Fam (2015)	Keshtegar et al. (2017)	RSM	ANN	GMDH-NN
Training phase	RMSE	3.1819	3.5701	3.1265	2.5758	2.5791	2.8925	2.3328	2.2279
	MSE	10.1243	12.7458	9.7752	6.6345	6.6516	8.3666	5.4419	4.9636
	MBE	-0.1196	0.2202	0.0215	0.1347	0.1553	0.1261	0.1130	0.1442
	CI	0.4653	0.3451	0.4721	0.6145	0.6158	0.5198	0.6824	0.7086
	Max (RE)	11.3958	13.1369	16.4585	16.1514	15.8321	14.6805	12.4658	10.6539
	min (RE)	-16.9798	-24.1197	-17.9425	-9.8271	-12.8366	-7.3668	-9.3270	-7.5183
Testing phase	RMSE	2.8793	3.3593	3.0762	2.5732	2.5204	2.8984	2.4899	2.1680
	MSE	8.2903	11.2851	9.4633	6.6216	6.3524	8.4008	6.1998	4.7000
	MBE	-0.0880	0.1904	0.0044	0.0980	0.1192	0.0980	0.0925	0.1126
	CI	0.5794	0.4487	0.5162	0.6390	0.6548	0.5475	0.6654	0.7419
	Max (RE)	7.6599	7.3586	9.2555	8.2535	8.1300	11.9600	9.3723	6.9405
	min (RE)	-11.9402	-15.1243	-13.0273	-6.7011	-6.9632	-7.1618	-4.6935	-3.4498
Over-all	RMSE	3.1214	3.5280	3.1165	2.5753	2.5673	2.8937	2.3642	2.2159
	MSE	9.7575	12.4537	9.7128	6.6319	6.5918	8.3734	5.5935	4.9109
	MBE	-0.1133	0.2143	0.0181	0.1273	0.1481	0.1205	0.1089	0.1379
	CI	0.4881	0.3658	0.4809	0.6194	0.6236	0.5253	0.6790	0.7153
	Max (RE)	10.6486	11.9812	15.0179	14.5719	14.2917	14.1364	11.8471	9.9112
	min (RE)	-15.9719	-22.3206	-16.9595	-9.2019	-11.6619	-7.3258	-8.4003	-6.7046

*The best outcomes are indicated in bold numbers.

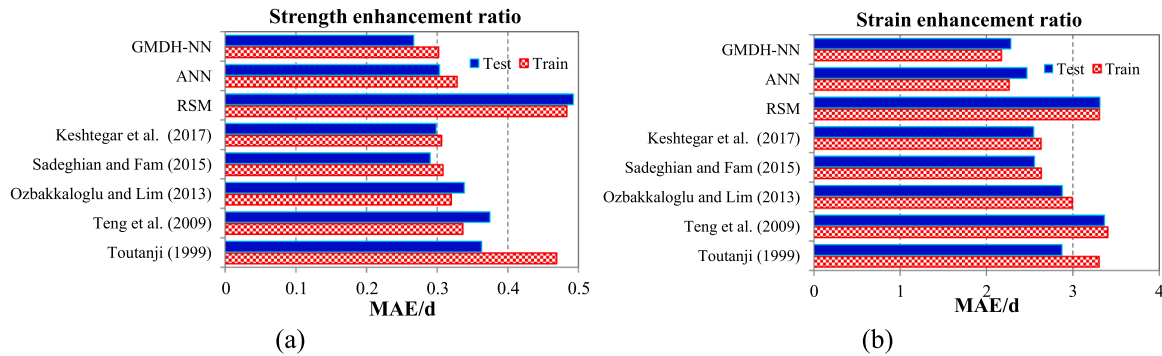


Fig. 8. Performance results of the proposed and studied models based on the MAE/d ratio for (a) the ultimate strength ratio (f_{cc}/f_{co}), and (b) the ultimate strain ratio ($\epsilon_{cc}/\epsilon_{co}$).

strain ratio ($\epsilon_{cc}/\epsilon_{co}$). Predicting the ultimate capacities of FRP-confined concrete using GMDH-NN produce the lowest RMSE, MSE values (i.e. in over-all), which is 0.2965 MPa and 0.0882 MPa for the ultimate strength ratio (f_{cc}/f_{co}), and 2.2159 MPa and 4.91 MPa for the ultimate strain ratio ($\epsilon_{cc}/\epsilon_{co}$), respectively. On the other hand, the GMDH-NN model indicates the highest CI values with $CI^{Ultimate\ strength} = 0.8198$ and $CI^{Ultimate\ strain} = 0.7153$. Furthermore, the GMDH-NN model provides the smallest differences in relative error ($|Max(RE) - Min(RE)|$) with values of 2.1 and 16.61 for the ultimate strength and strain ratios, respectively.

Among the empirical correlations, the Keshtegar et al. [16] model yielded the highest performance, with three best indicators, RMSE= 0.313 MPa, MSE= 0.098 MPa and CI= 0.799 for the ultimate strength ratio (f_{cc}/f_{co}), and RMSE= 2.57 MPa, MSE= 6.63 MPa and CI= 0.619 for the ultimate strain ratio ($\epsilon_{cc}/\epsilon_{co}$). Tables 2 and 3 inform as well that the RSM and Teng et al. [8] have the lowest performances when compared to all other models.

The MAE/d ratio is a helpful criterion for evaluating the effectiveness of a predictive model. A model's prediction outputs are more accurate and efficient if its MAE/d value is the lowest. Fig. 8 depicts the MAE/d values for the ultimate strength and strain ratios obtained using all the explored models in this study. Clearly, the GMDH-NN model produced the lowest MAE/d values for both, training and testing phases with MAE/d^{train}= 0.302 and MAE/d^{test}= 0.267 for predicting f_{cc}/f_{co} and MAE/d^{train}= 2.172 and MAE/d^{test}= 2.282 for predicting $\epsilon_{cc}/\epsilon_{co}$. Comparing the existed correlations performances, Keshtegar et al. [16] model performed the best with MAE/d^{train}= 0.306 and MAE/d^{test}= 0.298 for predicting f_{cc}/f_{co} and MAE/d^{train}= 2.63 and MAE/d^{test}= 2.544 for predicting $\epsilon_{cc}/\epsilon_{co}$. Among the other explored models, the ones with the highest MAE/d values are the RSM model (MAE/d^{train}= 0.484 and MAE/d^{test}= 0.493) and Toutanji et al. [9] model (MAE/d^{train}= 0.469 and MAE/d^{test}= 0.363) for predicting f_{cc}/f_{co} , and the RSM model (MAE/d^{train}= 3.306 and MAE/d^{test}= 3.311) and Teng et al. [8] model (MAE/d^{train}= 3.403 and MAE/d^{test}= 3.365) for predicting $\epsilon_{cc}/\epsilon_{co}$. Therefore, these models exhibit low accuracy, making them unsuitable for use as predictive models for the current issue and require refinement.

Using the testing phase, Figs. 9 and 10 analyze the agreement between predicted and experimental results. The coefficient of determination, R^2 , associated with the outcomes of the correlations or proposed machine learning models is presented in each sub-figure. The findings for the ultimate strength ratio (f_{cc}/f_{co}), shown in Fig. 9, reveal that the GMDH-NN outperforms all the models with an $R^2 = 0.871$. However, when compared to the existed empirical correlations, the two machine

learning approaches (ANN and RSM) have lower performance, with the RSM having the lowest agreement with the experimental data ($R^2 = 0.647$). For the ultimate strain ratio ($\epsilon_{cc}/\epsilon_{co}$), Fig. 10 illustrates that the proposed GMDH-NN model achieves the highest performance with an $R^2 = 0.79$, followed by the ANN ($R^2 = 0.728$) and Toutanji et al. [9] model ($R^2 = 0.728$). The Ozbakkaloglu and Lim [11] model ($R^2 = 0.638$) and the RMS technique ($R^2 = 0.631$) indicate the lowest performances. The lower performance the predictions in the ultimate strain capacity can be reflected by the associated data uncertainties as presented in Fig. 4 (Data description) for the test phase which capture a high-varieties of disperse data. As a result, the flexibility of the GMDH-NN model as a provided nonlinear function can improve prediction tendencies; however, in order to achieve better results, the impacts of input factors on the ultimate strain capacity should be considered. The highly nonlinear models, such as the GMDH-NN and ANN, make suitable predictions, but the failure mode under strain is influenced by complicated relationships that can be determined by input factors. The enhancement ratios in term of R^2 based on the GMDH-NN predictions to the best empirical correlation found to be 9.47% for the ultimate strain ratio ($\epsilon_{cc}/\epsilon_{co}$) and 4.31% for the ultimate strength ratio (f_{cc}/f_{co}). Thus, this graphical representation using scatter plots illustrates the effectiveness of the novel formulations by employing the proposed GMDH-NN in modeling process of this complex problem with high precision as compared to other machine learning models and existed correlations.

The Taylor diagram is used as a global visual indicator to demonstrate the performance of the suggested ML models and the empirical correlations. This graphical metric comprises three single measurements, namely standard deviation, person correlation coefficient, and RMSE. The closer the results are to the experimental observations, the better the model's overall performance relative to others. Fig. 11 depicts the Taylor diagrams using the testing results for the ultimate strain and strength capacity findings. These results revealed that the GMDH-NN model outperformed all other investigated models for predicting the ultimate capacity of FRP-confined concrete. The RSM, on the other hand, indicated the lowest performance when modeling the ultimate strength capacity, whereas Teng et al. [8] performed poorly when modeling the ultimate strain capacity.

5.2.2. Uncertainty evaluation

Finally, in Figs. 12 and 13, the prediction uncertainties during the modeling process are represented by the Experimental/Predicted (E/P) ratio for both ultimate capacities. These figures depict the training and testing phases, as well as the mean and standard deviation of the E/P

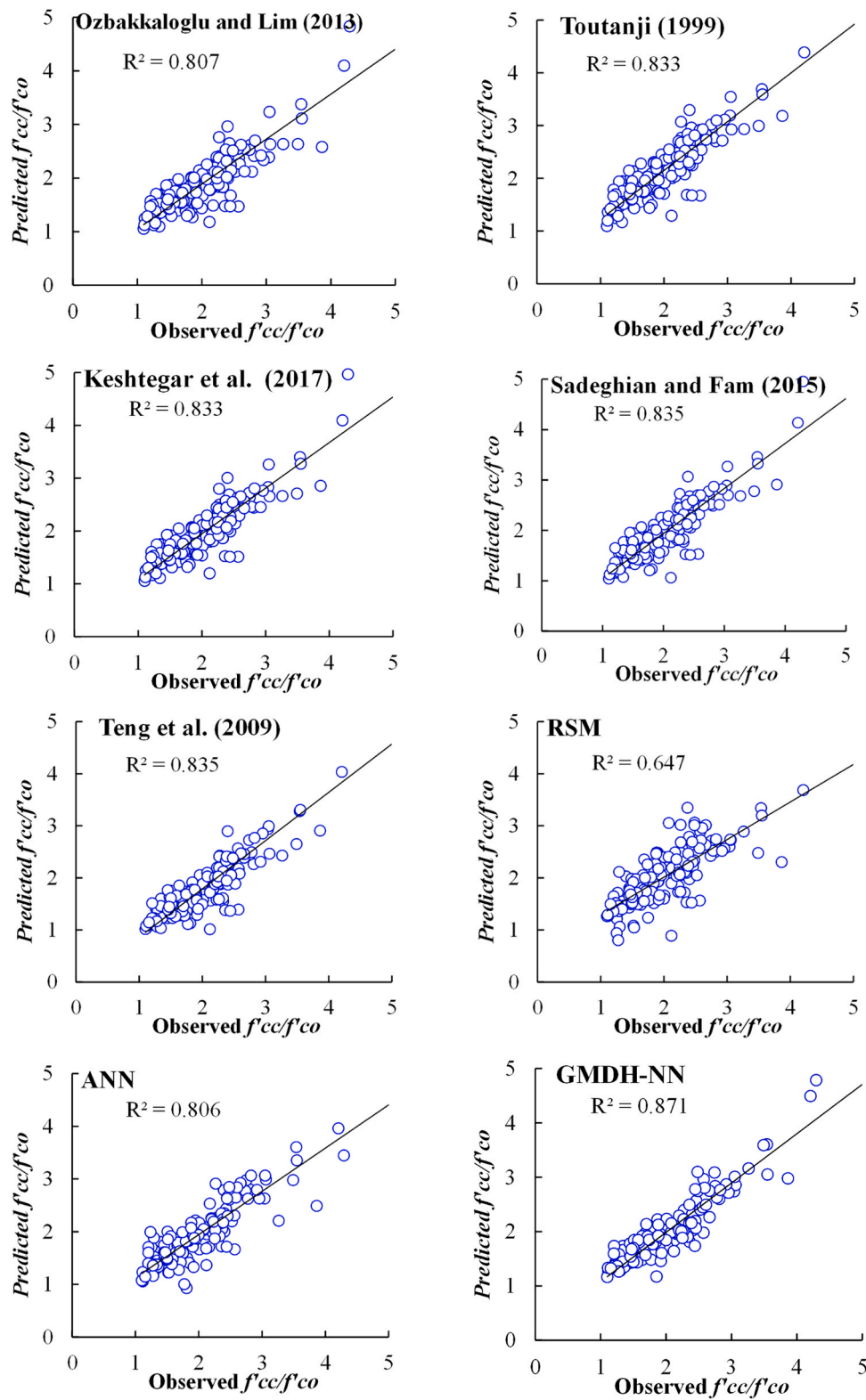


Fig. 9. Scatter plots of experimental tests versus predicted values for the ultimate strength ratio (f'_{cc}/f'_{co}) using the proposed machine learning models and empirical correlations.

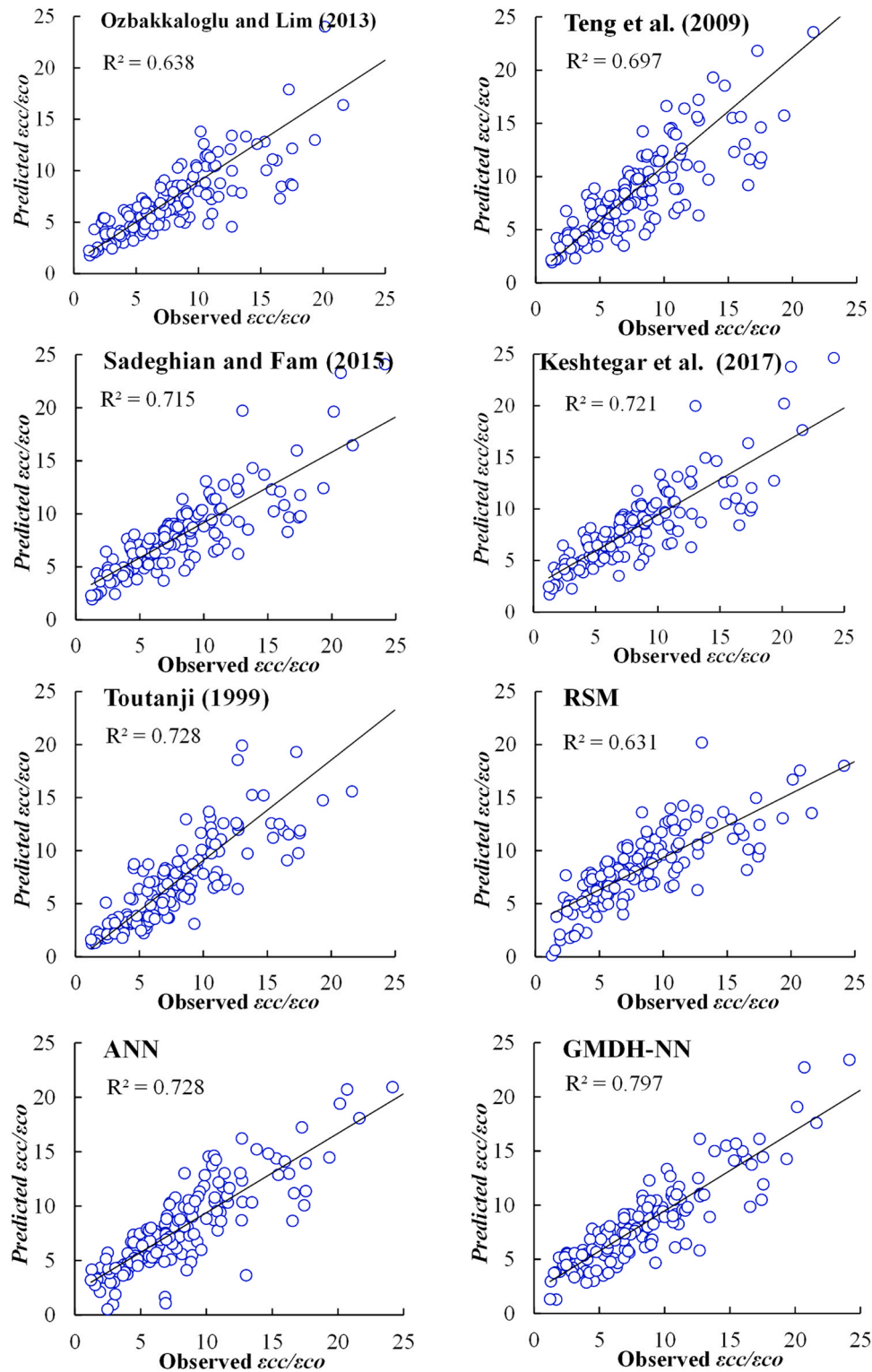


Fig. 10. Scatter plots of the experimental tests versus predicted values for the ultimate strain ratio ($\epsilon_{cc} / \epsilon_{co}$) using the proposed machine learning models and empirical correlations.

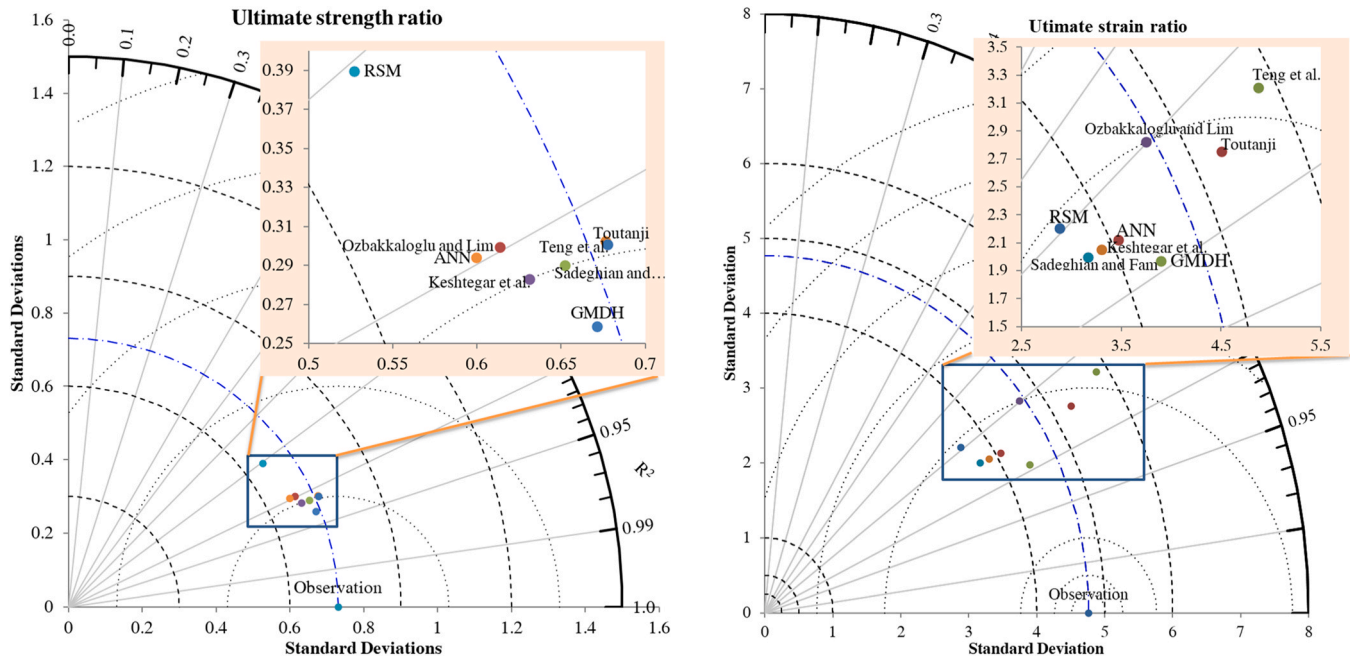


Fig. 11. Taylor diagram for evaluation of predicted test data by different models.

ratios (i.e. test phase) for each model. Noting that an accumulation of the results nearby the unit, indicates low uncertainties and excellent prediction abilities. Thus, according to these the figures the proposed GMDH-NN model is the most accurate for modeling the ultimate capacities of FRP-confined concrete, with mean and SD values for the ratio E/P equal to $Mean = 0.99$ and $STD = 0.21$ for the ultimate strength ratio (f_{cc}/f_{co}), and $Mean = 1$, $STD = 0.14$ for the ultimate strain ratio ($\epsilon_{cc}/\epsilon_{co}$). Teng et al. [8] ($Mean = 1.1$ and $STD = 0.19$) and Toutanji et al. [9] ($Mean = 0.91$ and $STD = 0.16$) have the highest uncertainties in predicting the ultimate strain capacity, whereas the RSM technique is considered as inaccurate to predict the ultimate strength ratio with a mean of 0.91 and STD of 1.15 .

5.3. Reliability analysis

Structural reliability analysis is a valuable probabilistic approach for investigating the safety levels of FRP-confined concrete structures by considering multi-source uncertainties such as sizes, material characteristics in concrete and FRP sheet composite core, and loads. To perform the reliability analysis, random variables are chosen depending on the inputs supplied in the previous modeling section, which are $D(mm)$, $t(mm)$, $f_c(MPa)$, $\epsilon_{h,rup}$, $E(GPa)$. The performance functions for approximating the failure domains are trained using the best obtainable model, which is the proposed GMDH-NN model. Noting that the key failure modes with uncertainty sources are the composite cores' strength and ductility. As a result, two performance functions for addressing the strength and strain failure modes of FRP-confined concrete segments are defined as follows:

$$g(f) = f_{cc}/f_{co} - 1.25 \quad (29)$$

$$g(\epsilon) = \epsilon_{cu}/\epsilon_{co} - 3 \quad (30)$$

where f_{cc}/f_{co} and $\epsilon_{cu}/\epsilon_{co}$ signify the ultimate strength and strain ratios as obtained using GMDH-NN model (i.e. formulations presented in Section

5.1.2). It is expected that the strength and ductility of unconfined concrete are increased by about 25% and about 200%, respectively, as indicated in the above performance functions. Furthermore, the statistical properties reported in Table 4 are utilized to analyze the structural reliability of three types of FRP sheets: CFRP, AFRP, and GFRP. In this work, the statistical features of FRP sheets, including mechanical properties and sizes, are determined using data analysis and probabilistic fitting of three distribution functions: Normal, Gumbel, and Lognormal. The concrete core strength (f_{co}) is modified from 30 to 100 (MPa) as a parameter for examining its implications in reliability evaluations.

The Monte Carlo Simulation (MCS) is used for performing the structural reliability analysis to estimate the failure probability of composite concrete cores. The MCS has a suitable accuracy and simplicity for approximating the failure probability, where the efficiency of MCS is enhanced based on the proposed advanced machine learning approach, GMDH-NN model [56,57]. For estimating the failure probabilities, three levels of the failure domain are given: 1) strength failure computed using Eqs. (29), 2) ductile failure computed using Eqs. (30), and 3) system failure computed using Eq. (29) and (30) as overall failure mode of $g = g(f) \cup g(\epsilon)$. MCS information may be found in Refs. [58–60] for reliability analysis with a total number of samples (NS) of 10^7 .

Using 10^7 samples, the histogram diagrams for the three performance functions evaluated by the ultimate capacities of CFRP, GFRP and AFRP-confined concrete are plotted in Figs. 14, 15 and 16, respectively. The resulting response utilizing the suggested GMDH-NN is captured differently for each performance function given by Eq. (29), (30), and $g = g(f) \cup g(\epsilon)$. The probability density functions (PDF) shown in Figs. 14–16 deviate considerably from the associated condition of the performance function for each FRP type (i.e. CFRP, GFRP, and AFRP). The PDFs for the system performance function are noted to have narrow bounds, as shown in Figs. a) 14–16, but the STDs of the simulated data generated higher values for the strain performance function, as shown in

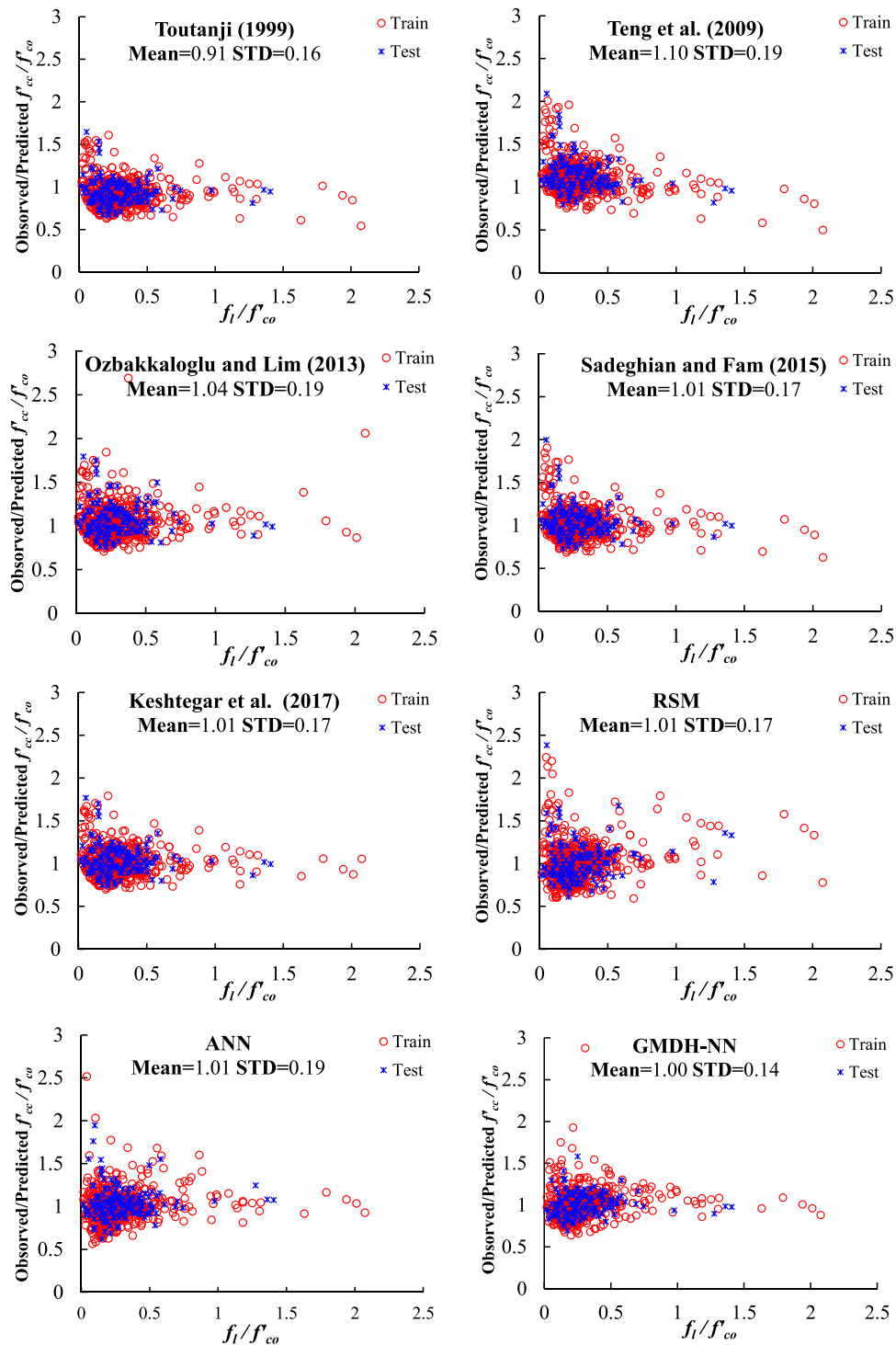


Fig. 12. Experimental to predicted ratio (E/P) for the ultimate strength ratio (f'_cc/f'_co) using the proposed machine learning models and empirical correlations.

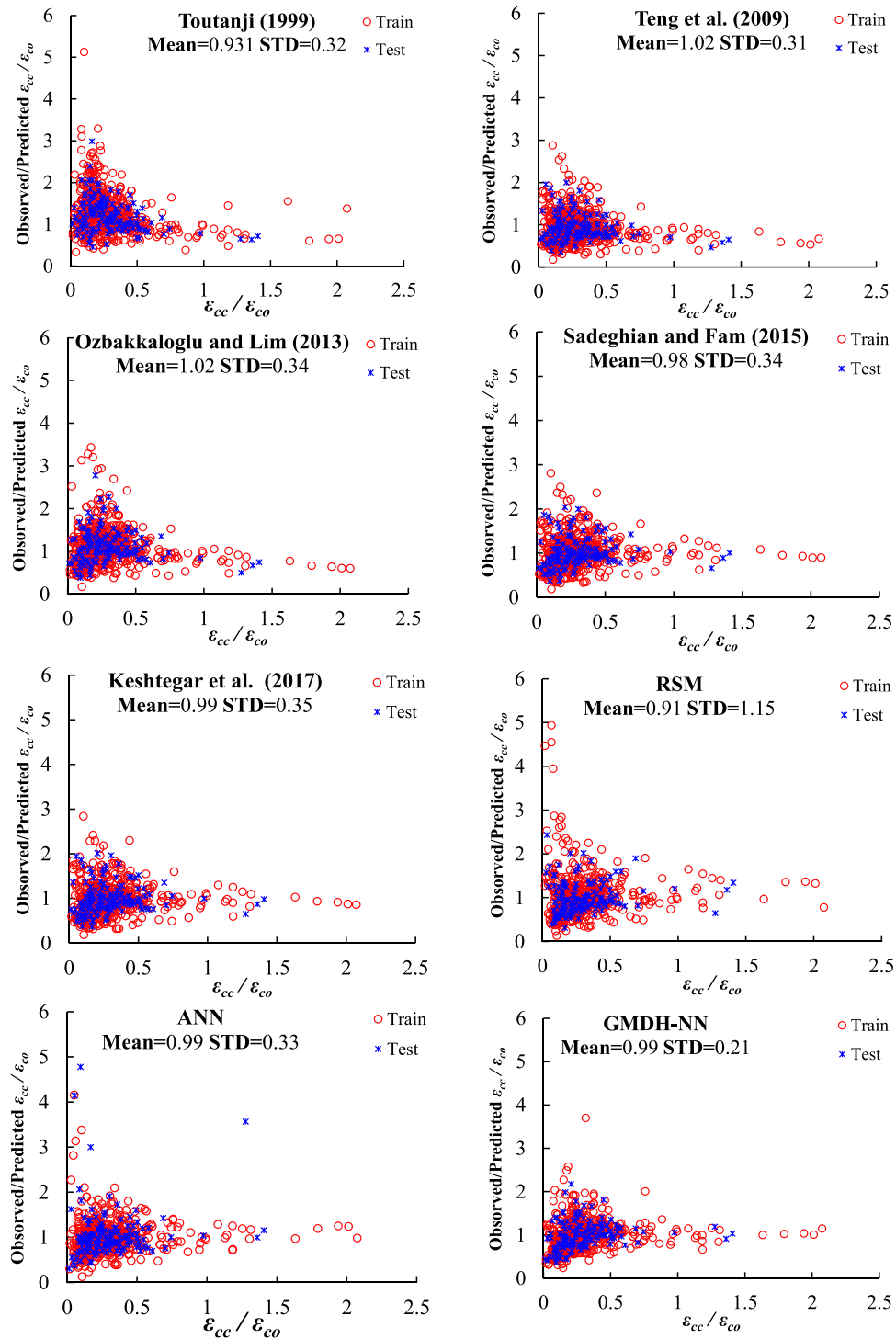


Fig. 13. Experimental to predicted ratio (E/P) for the ultimate strain ($\epsilon_{cc} / \epsilon_{co}$) ratio using the proposed machine learning models and empirical correlations.

Table 4

Statistical properties of the used random variables for FRP sheets and concrete cores.

Categories	Variable	Mean	COV	PDF
All FRP	f_{co} (MPa)	f_{co}	0.18	Normal
	D (mm)	150	0.1	Normal
CFRP	t (mm)	1.25	0.08	Lognormal
	$\varepsilon_{h,rup}$ (%)	1.01	0.352	Gumbel
	E (GPa)	185	0.1	Lognormal
GFRP	t (mm)	0.9	0.08	Gumbel
	$\varepsilon_{h,rup}$ (%)	3.12	0.391	Gumbel
	E (GPa)	85	0.1	Lognormal
AFRP	t (mm)	0.75	0.08	Lognormal
	$\varepsilon_{h,rup}$ (%)	2.62	0.218	Gumbel
	E (GPa)	120	0.1	Gumbel

Figs. c) 14–16. These figures show that the failure domain and data simulation for the three types of FRP sheets are determined using different PDFs, indicating that the FRP types have a significant impact

on the failure domain and reliability analyses of FRP-confined concrete based on strength and ductility capacities.

Fig. 17 depicts the failure probability for the three distinct failure domains given from the strain, strength, and system performance functions for the three different types of FRP warped sheets: a) CFRP, b) GFRP, and c) AFRP. It is worth noting that $f_{lu,a}/f'_{cc}$ values are computed depending on f'_{cc} , which may be changed up to 30 MPa. Using the mean values of the random variables reported in Table 4, the $f_{lu,a}$ values found as $f_{lu,a} = 31.1417$ MPa for CFRP, $f_{lu,a} = 31.8240$ MPa for GFRP and $f_{lu,a} = 31.44$ MPa for AFRP. Thus, $f_{lu,a}/f'_{cc}$ is reduced from 1.04 to 0.31, with the maximum value of $f_{lu,a}/f'_{cc}$ related to $f'_{cc} = 30$ MPa. Findings are plotted for the stress, strain, and system failure modes for CFRP, GFRP, and AFRP confined concrete, respectively. It is clear that the results of the system failure mode is always higher than the ones by the single failure modes, which is rational given that the failure probability of a system linked in series is higher than the failure probability of its components. According to the results of Fig. 17, the failure probabilities are evaluated differently for each type of FRP warped sheet. For the case

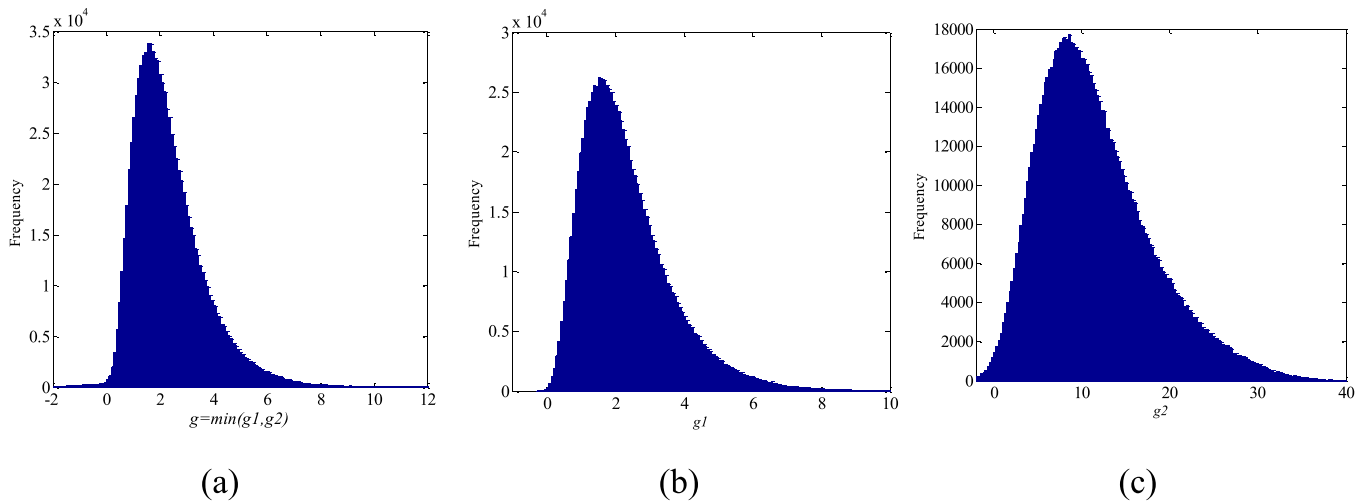


Fig. 14. Histogram of the three failure modes for CFRP warped sheet with $f'_{co} = 40$ MPa a) system performance function, b) strength performance function and c) ductility performance function.

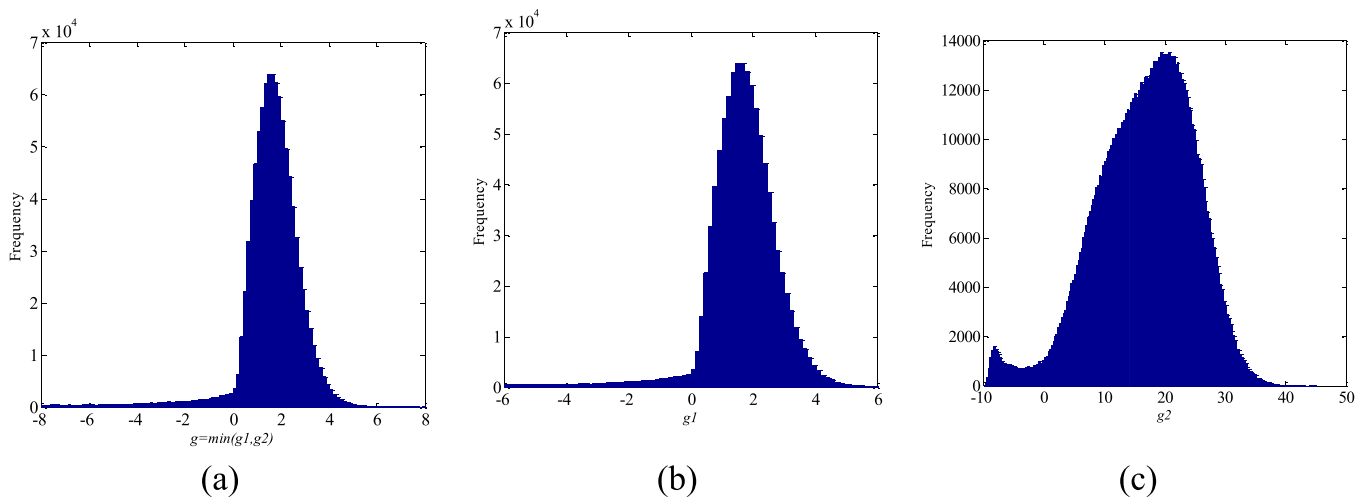


Fig. 15. Histogram of the three failure modes for GFRP warped sheet with $f'_{co} = 40$ MPa a) system performance function, b) strength performance function and c) ductility performance function.

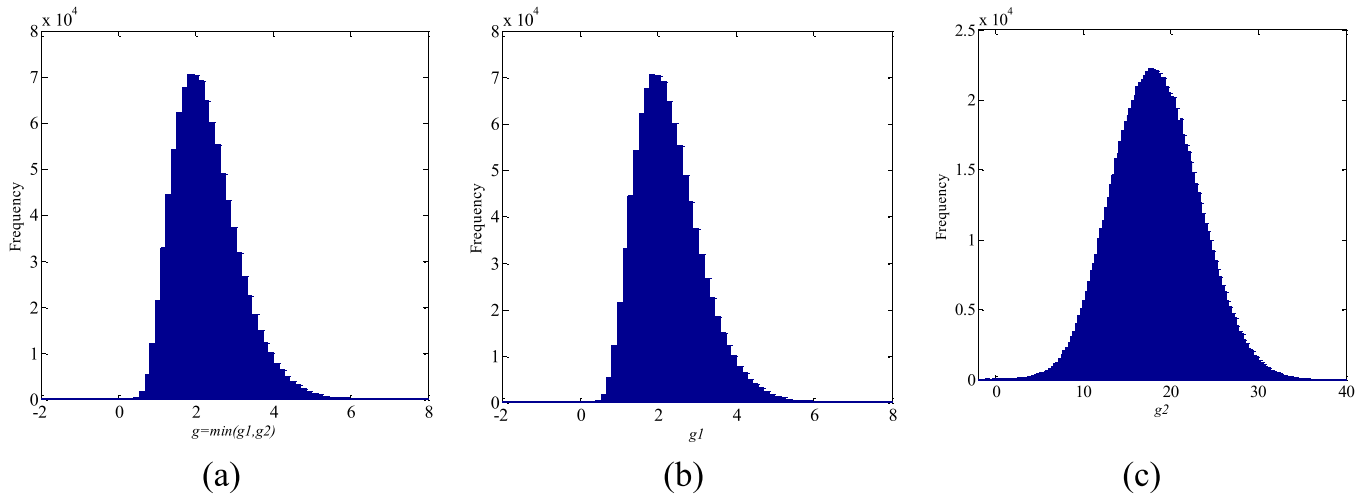


Fig. 16. Histogram of the three failure modes for AFRP warped sheet with $f_{co} = 40$ MPa a) system performance function, b) strength performance function and c) ductility performance function.

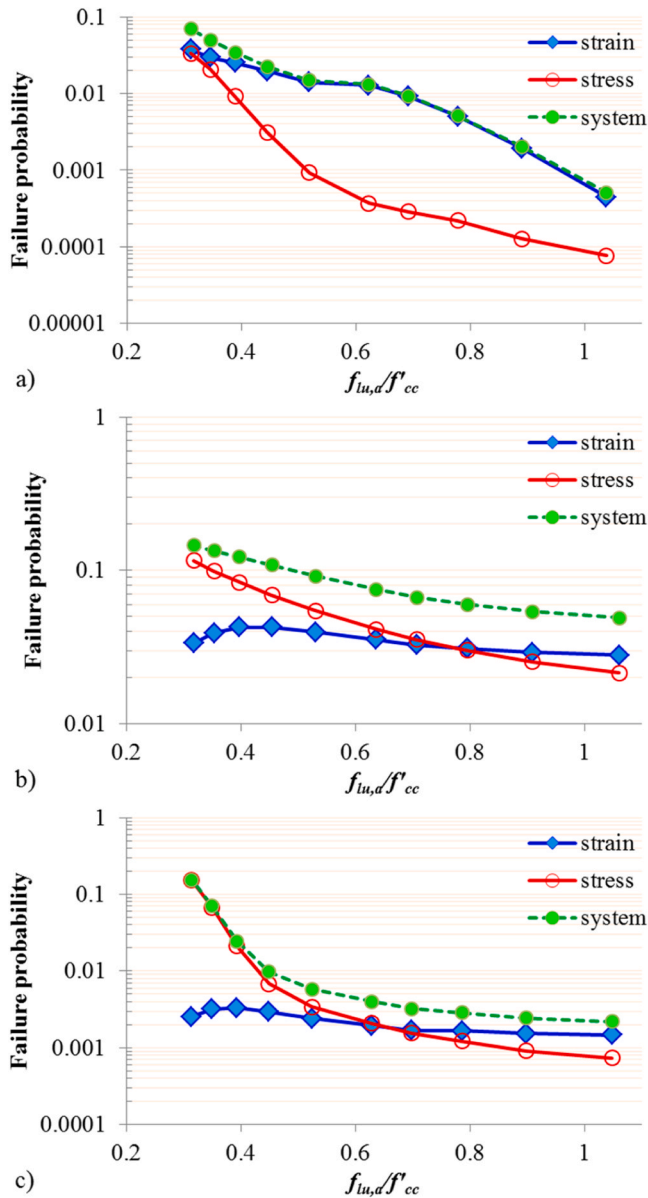


Fig. 17. Failure probability corresponding to $f_{lu,a}/f'_{cc}$ for different FRP warped sheets a) CFRP, b) GFRP and c) AFRP.

of CFRP confined concrete, the strain failure mode found to be dominant, however for the instances of GFRP and AFRP confined concrete, the strain failure mode was larger than the strength failure mode for $f_{lu,a}/f'_{cc} \geq 0.75$ and $f_{lu,a}/f'_{cc} \geq 0.65$, respectively. Another point to mention is that GFRP confined concrete has the highest failure rate when compared to other types of FRP-sheets, showing that such FRP-sheets have low enhancing characteristics. In this sense, CFRP sheets outperform other materials in terms of resistance to strength and ductility failure modes.

6. Conclusion

In this paper, we present a novel structural reliability-based machine learning framework for evaluating the failure probability of FRP-confined concrete with various sheet types. To that end, a novel ML technique, namely Group Method Data Handling Neural Network (GMDH-NN) is proposed for predicting the ultimate strain ($\epsilon_{cc}/\epsilon_{co}$) and strength (f_{cc}/f'_{co}) capacities of FRP-confined concrete. Through a large experimental database, the performance of the proposed correlations-based GMDH-NN is compared to that of current empirical correlations and other ML models, including artificial neural network (ANN) and response surface method (RSM). Following that, a structural reliability study is carried out using Monte Carlo simulation based on two performance functions (strength and ductility) merged with the generated GMDH-NN correlations. The main study findings are summarized as follows:

- The proposed GMDH-NN produces accurate closed form correlations for estimating the ultimate capacities of FRP-confined concrete. This unique approach differs from the traditional ML models (e.g., ANN and RSM), which are black boxes that the end user cannot reproduce.
- Using all evaluation metrics, the results indicate that GMDH-NN outperforms all the ML-based models and empirical correlations. GMDH-NN, in particular, produced R^2 values of 0.79 and 0.871 for $\epsilon_{cc}/\epsilon_{co}$ and f_{cc}/f'_{co} ratios, respectively. Thus, comparing to the best empirical correlation, predictions improved by 9.47% and 4.31% for $\epsilon_{cc}/\epsilon_{co}$ and f_{cc}/f'_{co} ratios, respectively.
- Keshtegar et al.'s model yielded acceptable results, whereas RSM was found to be unsuitable due to its low overall accuracy in dealing with the complicated database.
- Choosing the suitable FRP sheet is crucial for reducing failure probability, according to conducted structural reliability analysis. The results demonstrate that using CFRP sheets provides the highest level of safety when compared to other FRP sheets (GFRP and AFRP).

- In terms of strength and ductile failure modes, it was discovered that the failure probability of CFRP-confined concrete owing to the ductile failure mode (strain) is dominant, whereas it becomes dominant for GFRP and AFRP confined concrete when $f_{lu,a}/f_{cc} \geq 0.75$ and $f_{lu,a}/f_{cc} \geq 0.65$, respectively.

One of the obstacles that face the generalization of the presented framework is the dispersed data connected to the ultimate strain capacity, which requires further reinforcement in order to improve the prediction abilities of the proposed model. Furthermore, the collected data is limited to CFRP, GFRP, and AFRP sheets, which should be expanded to include other types of FRP sheets such as BFRP. Further research on reliability analysis approaches for FRP-confined columns should be conducted in the future.

CRediT authorship contribution statement

Mohamed El Amine Ben Seghier: Formal analysis, Investigation, Methodology, Project administration, Writing – original draft. **Behrooz Keshtegar:** Investigation, Methodology, Validation, Writing – original draft. **Tabatabaie Shojah:** Validation, Visualization, Writing – review & editing. **Mahmoud Alfouneh:** Data curation, Formal analysis, Investigation, Writing – original draft.

Declaration of Competing Interest

On behalf of all authors, we wish to confirm that there are no known conflicts of interest associated with this publication. No significant financial support has been made for the work that could have influenced its outcome.

Acknowledgements

The first, third and fourth authors want to acknowledge the support given by Iran National Foundation (INSF) under project No. 4012606 and University of Zabol (Grant No. IR-UOZ-2117).

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